

# A Planar Differential Inclusion with a Rotating Ellipsoidal Control Constraint

## Abstract

We describe a finite-dimensional benchmark problem for controlled differential inclusions. The example consists of a nonlinear time-dependent drift in  $\mathbb{R}^2$  and a set-valued control constraint given by a rotating, time-dependent ellipse. The admissible velocity set is represented as a Minkowski sum of the drift and the control ellipse. A smooth finite-dimensional parametrization of measurable selectors is introduced in such a way that the geometric constraint is satisfied by construction. This yields a numerically tractable optimal-control problem suitable for testing selector-based and neural parametrizations of differential inclusions.

## 1 Example

In this section we present a two-dimensional controlled differential inclusion with a nonautonomous drift and a rotating ellipsoidal set of admissible controls. The purpose of the example is twofold. First, it gives a transparent geometric model of a set-valued velocity field. Second, it provides a finite-dimensional numerical test problem in which admissibility of the selector is enforced by the parametrization itself, rather than through a penalty term.

Let  $T > 0$  and set  $J = [0, T]$ . In the numerical realization considered below we take

$$T = 3, \quad N = 120, \quad K = 24, \quad (1)$$

where  $N$  is the number of time-grid points and  $K$  is the number of control nodes used in the selector parametrization. The prescribed initial and target states are

$$x_0 = (0, 0)^\top, \quad x_1 = (1, -0.2)^\top. \quad (2)$$

The values of the selector-regularization parameter used in the continuation procedure are

$$\lambda_\xi \in \{10^{-3}, 10^{-2}, 10^{-1}\}. \quad (3)$$

We consider trajectories  $x : J \rightarrow \mathbb{R}^2$  satisfying the differential inclusion

$$\dot{x}(t) \in F(t, x(t)), \quad t \in J, \quad (4)$$

where the set-valued velocity field is of Minkowski type,

$$F(t, x) = g(t, x) + E(t). \quad (5)$$

Equivalently, an admissible trajectory may be written as the solution of

$$\dot{x}(t) = g(t, x(t)) + \xi(t), \quad \xi(t) \in E(t) \quad \text{for a.e. } t \in J. \quad (6)$$

Here  $g$  is the drift and  $\xi$  is a measurable selector of the time-dependent control set  $E(t)$ .

## 1.1 The drift field

The drift  $g : J \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is defined by

$$g(t, x) = \begin{pmatrix} \sin(x_1) + 0.15 \cos\left(\frac{2\pi t}{T}\right) \\ 0.5 \cos(x_2) - 0.20 \sin\left(\frac{2\pi t}{T}\right) \end{pmatrix}, \quad x = (x_1, x_2)^\top. \quad (7)$$

Thus the first component depends nonlinearly on  $x_1$ , the second component depends nonlinearly on  $x_2$ , and both components contain a periodic time forcing. In the geometric representation of the inclusion, the curve  $t \mapsto g(t, x(t))$  forms the centerline of the admissible velocity tube.

## 1.2 The rotating control ellipse

For  $t \in J$ , let

$$a(t) = 0.3 + 0.4 \left| \sin\left(\frac{\pi t}{T}\right) \right|, \quad b(t) = 0.15 + 0.25 \left| \cos\left(\frac{\pi t}{T}\right) \right|, \quad (8)$$

and define the rotation angle

$$\theta(t) = \frac{\pi t}{T}. \quad (9)$$

The rotation matrix is denoted by

$$R_{\theta(t)} = \begin{pmatrix} \cos \theta(t) & -\sin \theta(t) \\ \sin \theta(t) & \cos \theta(t) \end{pmatrix}. \quad (10)$$

The admissible selector set is the ellipse

$$E(t) = \left\{ R_{\theta(t)} \begin{pmatrix} u \\ w \end{pmatrix} \in \mathbb{R}^2 : \frac{u^2}{a(t)^2} + \frac{w^2}{b(t)^2} \leq 1 \right\}. \quad (11)$$

Since  $\theta(0) = 0$  and  $\theta(T) = \pi$ , the ellipse performs a half-turn over the interval  $J$ . Consequently, the family of translated sets  $g(t, x(t)) + E(t)$  forms a twisted tube of admissible velocities along the drift curve.

For  $\xi = (\xi_1, \xi_2)^\top \in \mathbb{R}^2$ , the coordinates in the local frame of the ellipse are given by

$$\begin{pmatrix} u(t) \\ w(t) \end{pmatrix} = R_{-\theta(t)} \xi(t), \quad (12)$$

that is,

$$u(t) = \cos \theta(t) \xi_1(t) + \sin \theta(t) \xi_2(t), \quad w(t) = -\sin \theta(t) \xi_1(t) + \cos \theta(t) \xi_2(t). \quad (13)$$

It is convenient to introduce the ellipse-level function

$$\ell(t, \xi) = \left( \frac{u(t)}{a(t)} \right)^2 + \left( \frac{w(t)}{b(t)} \right)^2. \quad (14)$$

Then  $\xi(t) \in E(t)$  if and only if  $\ell(t, \xi(t)) \leq 1$ .

### 1.3 Selector parametrization

The selector is not optimized independently at all grid points. Instead, it is generated from  $K$  control nodes and then interpolated to the full time grid. This reduces artificial oscillations and gives a low-dimensional representation of the admissible selector.

Let

$$w_c^j \in \mathbb{R}^2, \quad \rho_c^j \in \mathbb{R}, \quad j = 0, \dots, K-1, \quad (15)$$

be the nodal variables. Linear interpolation of these nodes yields functions  $w(t_i) \in \mathbb{R}^2$  and  $\rho(t_i) \in \mathbb{R}$  on the grid

$$t_i = i\Delta t, \quad i = 0, \dots, N-1, \quad \Delta t = \frac{T}{N-1}. \quad (16)$$

For a small parameter  $\varepsilon > 0$ , taken numerically as  $\varepsilon = 10^{-10}$ , define

$$d(t_i) = \frac{w(t_i)}{\sqrt{\|w(t_i)\|^2 + \varepsilon}}. \quad (17)$$

The radial magnitude is generated by the sigmoid function

$$s(t_i) = \sigma(\rho(t_i)) = \frac{1}{1 + e^{-\rho(t_i)}}. \quad (18)$$

Therefore  $0 < s(t_i) < 1$ . The local coordinates of the selector are set to be

$$u(t_i) = s(t_i)a(t_i)d_1(t_i), \quad w(t_i) = s(t_i)b(t_i)d_2(t_i), \quad (19)$$

and the selector itself is defined by

$$\xi(t_i) = R_{\theta(t_i)} \begin{pmatrix} s(t_i)a(t_i)d_1(t_i) \\ s(t_i)b(t_i)d_2(t_i) \end{pmatrix}. \quad (20)$$

Equivalently,

$$\xi_1(t_i) = s(t_i) [\cos \theta(t_i)a(t_i)d_1(t_i) - \sin \theta(t_i)b(t_i)d_2(t_i)], \quad (21)$$

$$\xi_2(t_i) = s(t_i) [\sin \theta(t_i)a(t_i)d_1(t_i) + \cos \theta(t_i)b(t_i)d_2(t_i)]. \quad (22)$$

**Proposition 1.1** (Admissibility by construction). *For every grid point  $t_i$ , the selector defined by (20) satisfies*

$$\xi(t_i) \in E(t_i). \quad (23)$$

More precisely,

$$\ell(t_i, \xi(t_i)) = s(t_i)^2 (d_1(t_i)^2 + d_2(t_i)^2) \leq s(t_i)^2 < 1, \quad (24)$$

up to the harmless normalization regularization introduced by  $\varepsilon$ .

*Proof.* By construction, the local coordinates of  $\xi(t_i)$  are precisely those in (19). Substituting them into (14) gives

$$\ell(t_i, \xi(t_i)) = \left( \frac{s(t_i)a(t_i)d_1(t_i)}{a(t_i)} \right)^2 + \left( \frac{s(t_i)b(t_i)d_2(t_i)}{b(t_i)} \right)^2. \quad (25)$$

This yields (24). Since  $0 < s(t_i) < 1$  and  $\|d(t_i)\| \leq 1$ , the selector lies in the interior of  $E(t_i)$ .  $\square$

*Remark 1.2.* The constraint  $\xi(t_i) \in E(t_i)$  is therefore not imposed by adding a penalty for constraint violation. It is built into the parametrization. This is useful in numerical methods for differential inclusions because the residual relation

$$\dot{x}(t) - g(t, x(t)) \in E(t) \quad (26)$$

can be replaced by the equality

$$\dot{x}(t) - g(t, x(t)) = \xi(t), \quad \xi(t) \in E(t), \quad (27)$$

where the second condition is automatically satisfied.

## 1.4 State dynamics and time integration

For a fixed selector  $\xi$ , the state equation is the ordinary differential equation

$$\dot{x}(t) = g(t, x(t)) + \xi(t), \quad x(0) = x_0. \quad (28)$$

The numerical goal is to find an admissible selector for which

$$x(T) \approx x_1. \quad (29)$$

The dynamics are integrated on the grid (16) by the classical fourth-order Runge–Kutta method. Writing

$$f(t, x) = g(t, x) + \xi(t), \quad (30)$$

one step of the scheme is

$$\begin{aligned} k_1 &= f(t_k, x_k), \\ k_2 &= f\left(t_k + \frac{\Delta t}{2}, x_k + \frac{\Delta t}{2}k_1\right), \\ k_3 &= f\left(t_k + \frac{\Delta t}{2}, x_k + \frac{\Delta t}{2}k_2\right), \\ k_4 &= f(t_k + \Delta t, x_k + \Delta tk_3), \end{aligned}$$

and

$$x_{k+1} = x_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4). \quad (31)$$

Between grid points, the selector is evaluated by linear interpolation.

## 1.5 Finite-dimensional optimal-control problem

The optimization variable is

$$z = (w_c^0, \dots, w_c^{K-1}, \rho_c^0, \dots, \rho_c^{K-1}) \in \mathbb{R}^{3K}. \quad (32)$$

For  $K = 24$  this gives  $3K = 72$  scalar variables. For each  $z$ , the selector  $\xi_z(t_i)$  is reconstructed using (20); the state  $x_z(t_i)$  is obtained from (31); and the selected velocity is

$$v_z(t_i) = g(t_i, x_z(t_i)) + \xi_z(t_i). \quad (33)$$

The cost functional is taken in the form

$$J(z) = \lambda_T J_T(z) + \lambda_\xi J_\xi(z) + \lambda_s J_s(z) + \lambda_c J_c(z) + \lambda_b J_b(z) + \lambda_u J_u(z). \quad (34)$$

The individual terms are defined as follows:

$$J_T(z) = \|x_z(T) - x_1\|^2, \quad (35)$$

$$J_\xi(z) = \frac{1}{N} \sum_{i=0}^{N-1} \ell(t_i, \xi_z(t_i)), \quad (36)$$

$$J_s(z) = \frac{1}{N-1} \sum_{i=0}^{N-2} \|\xi_z(t_{i+1}) - \xi_z(t_i)\|^2, \quad (37)$$

$$J_c(z) = \frac{1}{N-2} \sum_{i=1}^{N-2} \|\xi_z(t_{i+1}) - 2\xi_z(t_i) + \xi_z(t_{i-1})\|^2, \quad (38)$$

$$J_b(z) = \frac{1}{N} \sum_{i=0}^{N-1} \ell(t_i, \xi_z(t_i))^3, \quad (39)$$

$$J_u(z) = \frac{1}{K-1} \sum_{j=0}^{K-2} \|w_c^{j+1} - w_c^j\|^2 + \frac{1}{K-1} \sum_{j=0}^{K-2} (\rho_c^{j+1} - \rho_c^j)^2. \quad (40)$$

The terminal weight is chosen as

$$\lambda_T = 2500. \quad (41)$$

The term  $J_T$  enforces closeness to the prescribed terminal state. The term  $J_\xi$  discourages selectors located near the boundary of the ellipse. The terms  $J_s$  and  $J_c$  control first- and second-order discrete variations of the selector, while  $J_b$  acts as an additional soft barrier against boundary saturation. Finally,  $J_u$  regularizes the nodal variables before interpolation.

With the weights used in the implementation, the problem may be written explicitly as

$$\begin{aligned} \min_{z \in \mathbb{R}^{72}} \quad & 2500 \|x_z(T) - x_1\|^2 + \lambda_\xi \frac{1}{N} \sum_{i=0}^{N-1} \ell(t_i, \xi_z(t_i)) \\ & + \frac{1}{N-1} \sum_{i=0}^{N-2} \|\xi_z(t_{i+1}) - \xi_z(t_i)\|^2 \\ & + 0.2 \frac{1}{N-2} \sum_{i=1}^{N-2} \|\xi_z(t_{i+1}) - 2\xi_z(t_i) + \xi_z(t_{i-1})\|^2 \\ & + 10^{-2} \frac{1}{N} \sum_{i=0}^{N-1} \ell(t_i, \xi_z(t_i))^3 \\ & + 5 \cdot 10^{-2} \left[ \frac{1}{K-1} \sum_{j=0}^{K-2} \|w_c^{j+1} - w_c^j\|^2 + \frac{1}{K-1} \sum_{j=0}^{K-2} (\rho_c^{j+1} - \rho_c^j)^2 \right], \end{aligned} \quad (42)$$

subject to

$$\dot{x}_z(t) = g(t, x_z(t)) + \xi_z(t), \quad x_z(0) = x_0, \quad (43)$$

and the built-in admissibility condition

$$\xi_z(t_i) \in E(t_i), \quad i = 0, \dots, N-1. \quad (44)$$

The optimization is performed by the L-BFGS-B method, using warm starts along the sequence of regularization parameters  $\lambda_\xi \in \{10^{-3}, 10^{-2}, 10^{-1}\}$ .

## 1.6 Geometric interpretation

At each time  $t_i$ , the translated ellipse

$$g(t_i, x_z(t_i)) + E(t_i) \quad (45)$$

represents the set of all admissible velocities of the inclusion. Plotting these ellipses for all grid points produces a twisted tube in the space of coordinates (component 1, component 2,  $t$ ). The centerline of this tube is the drift curve  $g(t_i, x_z(t_i))$ , while the selected velocity is

$$v_z(t_i) = g(t_i, x_z(t_i)) + \xi_z(t_i). \quad (46)$$

Thus each vector  $\xi_z(t_i)$  is the displacement from the center of the admissible velocity set to the actually selected velocity. Figure 1 shows this geometry: the drift curve forms the centerline, while the sampled velocities lie inside the translated rotating ellipses.

Equivalently, in the coordinate space  $(\xi_1, \xi_2, t)$  one observes the selector itself inside the time-dependent ellipse  $E(t)$  centered at the origin. The value of the level function  $\ell(t_i, \xi_z(t_i))$  measures the normalized radial position of the selector inside the corresponding ellipse; this is illustrated in Figure 2.

The corresponding state trajectory is shown in Figure 3. It starts at  $x_0$  and approaches the prescribed terminal profile  $x_1$  under the selected admissible velocities.

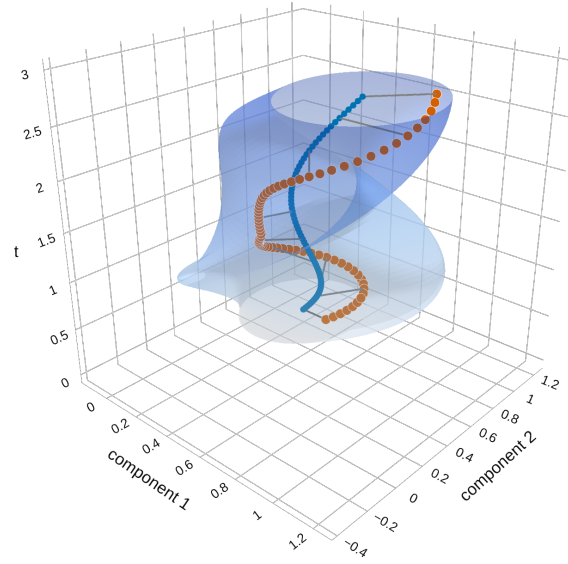
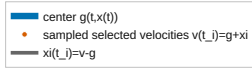


Figure 1: Admissible velocity tube associated with the inclusion  $\dot{x}(t) \in g(t, x(t)) + E(t)$ . The blue curve represents the center  $g(t, x(t))$ , the orange points represent sampled selected velocities  $v(t_i) = g(t_i, x(t_i)) + \xi(t_i)$ , and the grey segments indicate the selector displacement  $\xi(t_i) = v(t_i) - g(t_i, x(t_i))$ .

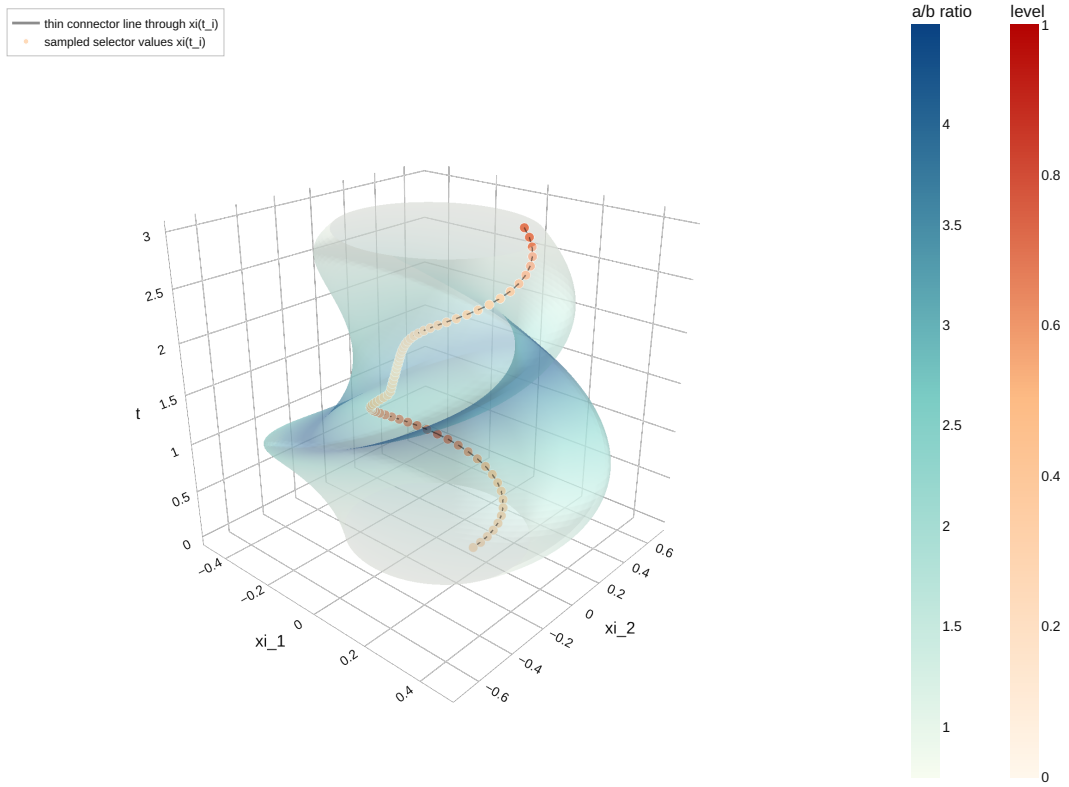


Figure 2: Selector trajectory inside the rotating control cylinder. The surface represents the family of admissible ellipses  $E(t)$ , while the sampled points show  $\xi(t_i)$ . The color scale associated with the points represents the level  $\ell(t_i, \xi(t_i))$ , and the surface coloring indicates the time-dependent anisotropy ratio  $a(t)/b(t)$ .

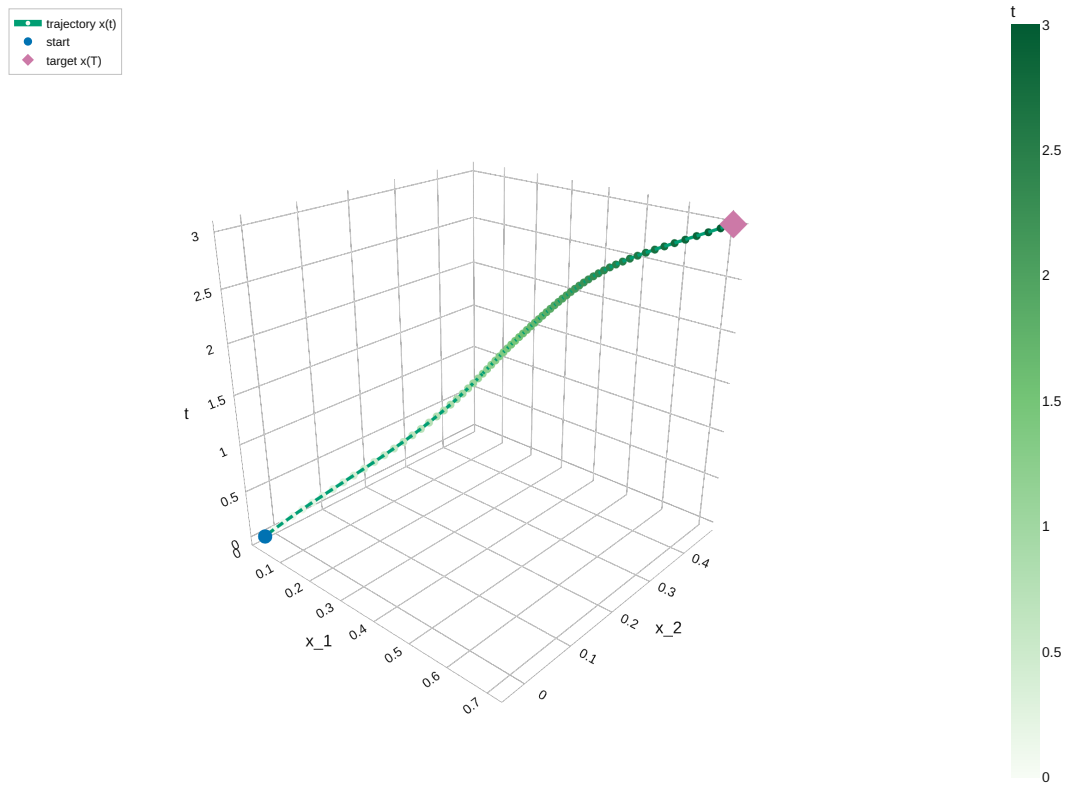


Figure 3: State trajectory  $x(t)$  generated by the optimized admissible selector. The trajectory starts from  $x_0 = (0, 0)^\top$  and is steered toward the target state  $x_1 = (1, -0.2)^\top$  over the interval  $[0, T]$ .

*Remark 1.3* (Relation to selector-based methods). This example can be viewed as a prototype for numerical methods for controlled differential inclusions. In the language of control theory,  $\xi$  is a geometrically constrained control. In the language of differential inclusions,  $\xi$  is a measurable selection of the set-valued map  $t \mapsto E(t)$ . In neural or physics-informed parametrizations, the same construction suggests replacing a penalty-based treatment of the inclusion constraint by an internal parametrization of admissible selectors.

**Theorem 1.4** (Numerical feasibility of the selector formulation). *Every discrete selector generated by (20) yields a discrete trajectory satisfying the inclusion constraint at the grid level, namely*

$$\frac{x_{i+1} - x_i}{\Delta t} \in g(t_i, x_i) + E(t_i) \quad (47)$$

*up to the consistency error of the chosen time-integration scheme and the interpolation of  $\xi$ .*

*Proof.* By Proposition 1.1,  $\xi(t_i) \in E(t_i)$  for every grid point. The right-hand side used in the time integrator has the form  $g(t_i, x_i) + \xi(t_i)$ , and therefore belongs to  $g(t_i, x_i) + E(t_i)$ . The stated assertion follows after accounting for the standard local truncation error of the Runge–Kutta discretization and the interpolation used to evaluate  $\xi$  between grid points.  $\square$

In summary, the example represents the controlled inclusion

$$\dot{x}(t) \in g(t, x(t)) + E(t) \quad (48)$$

by a smooth parametrized selector  $\xi(t) \in E(t)$  and solves the finite-dimensional problem of choosing this selector so that the trajectory starting from  $x_0$  reaches, as closely as possible, the prescribed target  $x_1$  at time  $T$ .